

EE5203 L01 Practice Worksheet

From C code to a blinking LED: what each step proves - Basri Erdogan

Time: 30-45 minutes

Rule: Try every question before you open the solutions. Short answers are fine. One sentence is usually enough.

Part A - Name the parts (5 minutes)

1. Write one word from this list next to each item: **processor, microcontroller, peripheral, board, embedded system.**

Item	Your word
Arm Cortex-M4	
STM32F401RE	
Nucleo-F401RE	
TIM2 (a timer inside the chip)	
A thermostat: STM32 board + temperature sensor + relay	

2. An embedded system has limits. Give one example for each limit. Example for time: *“it must react within 10 ms”*.
- a. Memory: ...
 - b. Energy: ...
 - c. Safety: ...
3. The console says **“Build: 0 errors”**, but the LED does not blink. Is this possible? **Yes / No**. Explain in one sentence.

Part B - The path from code to LED (10 minutes)

4. Put the six steps in order. Write 1 (first) to 6 (last).

Step	Order
Flash the firmware into the chip	
Link the object files into one program	
Edit <code>main.c</code>	
Compile the C files	
Watch the LED	
Look at a variable at a breakpoint	

5. Each message comes from one step of the path. Match them. Steps: **compile, link, flash, run.**

Message or symptom	Step
expected ';' before '}'	
undefined reference to process_sample	
No ST-LINK detected	
The program stops at the breakpoint, but the variable has the wrong value	

6. Each sentence below is a claim. Write **what you would look at** to prove it: the build console, the download message, the debugger, or the board itself.
- “My code compiled.”
 - “The program is inside the chip.”
 - “The loop runs again and again.”
 - “The LED blinks once per second.”

Part C - Predict before you run (10 minutes)

7. Read the code. Do **not** run it.

```
1 int count = 0;
2
3 while (count < 3) {
4     count = count + 1;
5 }
```

- a. Fill the table.

After loop pass	count is
1	
2	
3	

- b. What is the final value of `count`?
- c. On which line would you put a breakpoint to watch `count` change? Why there?
8. Two programs. Only one number is different.

<pre>/* Program A */ while (1) { HAL_GPIO_TogglePin(...); HAL_Delay(500); }</pre>	<pre>/* Program B */ while (1) { HAL_GPIO_TogglePin(...); HAL_Delay(100); }</pre>
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- a. In 10 seconds, about how many times does the LED toggle in Program A? In Program B?

- b. What stays the same in both programs?
- c. How would you check your answer with a phone stopwatch?
9. You change 500 to 100, press **F5**, and the LED blinks exactly as before. Give **three** possible reasons. For each reason, write one quick test.

Possible reason	Quick test
1.	
2.	
3.	

Part D - Ask for help the right way (10 minutes)

10. A student says: “*My code does not work.*” Here are the facts: the build and the download succeeded; LD2 should blink every 0.5 s; LD2 stays off; a breakpoint inside `while (1)` is never hit. Write the four lines:
 - **Expected:** ...
 - **Observed:** ...
 - **Evidence:** ...
 - **Next test:** ...
11. A breakpoint on `HAL_GPIO_TogglePin(...)` is never reached. Number these checks from 1 (do first) to 5 (do last). Then explain your number 1 in one sentence.

Check	Order
Replace the USB cable	
Check that the program reaches <code>main()</code>	
Check that the correct project was flashed	
Check whether the program is stopped somewhere else	
Check the loop and <code>if</code> conditions	

12. The LED blinks correctly. In the debugger, `toggle_count` still shows 0 after 10 seconds. Can the blinking LED prove that `toggle_count` is correct? **Yes / No**. Give two reasons.

Part E - Your project idea (5 minutes)

13. Choose one device you know: an alarm clock, a kettle, a bike light, a parking sensor, anything. Fill the five boxes with one word or a short phrase each.

Input	Acquisition	Processing	Output	Communication

14. Describe the **smallest version that works** in at most three sentences. Then name **one extra feature** you would add only after the small version works.

Self-check

Mark one:

- ☐ I am ready for Lab 1.
- ☐ I need to review the path from code to LED (Part B).
- ☐ I need help with C loops and variables (Part C).
- ☐ I need help with the VS Code / CubeMX setup.